LESSON 3.1

Understand Ratios

Start here · Words, worked example, and sentence starters

Name _____ **Date** _____ **Period** _____

LEARNING TARGET I can write and describe a ratio that compares two quantities.

WHY THIS MATTERS Welcome to Chef Academy! Head Chef Reyes is teaching new students how to read recipes like a pro. Her sparkling cranberry-apple fruit drink calls for 3 cups of apple juice for every 2 cups of sparkling water. Before the students can mix the drink, they need to understand how these ingredient amounts compare to each other.

START HERE

Words for this lesson

Read each word before you begin. Say it out loud.

WORD	WHAT IT MEANS	EXAMPLE
Ratio *Spanish: Razón*	A way to compare two amounts, like 3 to 2.	3 to 5 ■■■ □□□□□
Comparison *Spanish: Comparación*	Looking at two or more amounts to see how they are related.	5 eggs vs. 3 eggs — which recipe uses more?
Part-to-part *Spanish: Parte a parte*	A ratio comparing one part of a group to another part.	apples to oranges
Part-to-whole *Spanish: Parte a todo*	A ratio comparing one part to the whole group.	apples to all the fruit
Colon notation *Spanish: Notación con dos puntos*	Writing a ratio with two dots between the numbers, like 3:2.	3 to 2 can be written as 3:2
Tape diagram *Spanish: Diagrama de cintas*	A bar cut into equal parts that shows how two amounts in a ratio compare, and how they make up the whole.	3 equal parts of apple juice beside 2 equal parts of sparkling water — 5 parts of fruit drink in all.

HOW IT WORKS

Worked example

These numbers are not on your problems. The steps are. Follow them with your own numbers.

A shelf has 6 math books and 9 reading books. Write reading to math, then math to all books.

1. Reading first: 9 : 6.
2. All books: 6 + 9 = 15, so math to all is 6 : 15.
3. Divide both terms by 3: 6 : 15 = 2 : 5.

Answer: 9 : 6 and 2 : 5

NOW YOU TRY

Same steps, your turn

The steps are the same as the worked example. The numbers are yours. Do the work.

A bowl has 4 apples and 1 orange. What two amounts are we comparing?

1. _____
2. _____
3. _____
4. _____
5. _____

Answer: _____

SAY IT AND WRITE IT

Sentence starters

Finish each sentence out loud with a partner. Then use them in your writing.

- The ratio of _____ to _____ is _____ .
- This means for every _____ _____ , there are _____ _____ .
- The order matters because _____ .
- I simplified by dividing both terms by _____ .

Word bank ratio | term | compare | for every | part | whole | total | simplest form | divide | order | first | second

WATCH OUT!

A common mistake

A common mistake in Understand Ratios is writing the ratio in the wrong order: when asked for the ratio of apple juice to sparkling water in a recipe that uses 3 cups of apple juice for every 2 cups of sparkling water, some students write 2:3 (sparkling water to apple juice) because they list the ingredients in whatever order comes to mind rather than the order the question asks for. The two ratios describe different drinks

LESSON 3.1

Understand Ratios

Version A · Supported practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each statement: does it describe a ratio or NOT describe a ratio?

Groups: IS a Ratio | NOT a Ratio

_____ 7 pencils for every 3 erasers

_____ 10 red beads to 4 blue beads

_____ There are 15 chairs in the room

_____ The pizza was delicious

HINT A ratio must COMPARE two amounts. Ask each card: does it mention two quantities, or just one?

PART 2 Apply it

real-world problems

2 **Circle the letter of the best answer. Show how you know.**

A recipe uses 4 cups of milk and 1 cup of cream. What is the ratio of milk to cream?

(A) 1:4 (B) 4:1

(C) 4:5 (D) 5:1

HINT Re-read the question: it asks for milk to cream, in that exact order.

SHOW YOUR WORK

3 **Circle the letter of the best answer. Show how you know.**

Chef Reyes uses 6 strawberries and 2 bananas in a smoothie. What is the part-to-whole ratio of bananas to total fruit?

(A) 2:8
(B) 2:6
(C) 6:2
(D) 8:2

HINT Notice the word 'total' — this is a part-to-WHOLE ratio, not banana-to-strawberry.

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

In a class of 30 students, 18 are girls. What is the part-to-part ratio of boys to girls?

(A) 12:18
(B) 12:30
(C) 18:12
(D) 18:30

HINT The problem tells you girls (18) and the whole class (30) — but it asks about BOYS to girls.

SHOW YOUR WORK

5 **Circle the letter of the best answer. Show how you know.**

A pet store has 5 dogs and 9 cats. What is the ratio of dogs to cats?

(A) 5:9
(B) 5:14
(C) 9:5
(D) 14:5

HINT Re-read: the question asks for dogs to cats, in that order.

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

6 Find the mistake. Explain it, then show the correct work.

Spot the Mistake — Order of a Ratio

1. **Read the problem** A salad bar has 5 bowls of soup and 8 sandwiches. Write the ratio of soup to sandwiches.
2. **Identify the amounts** Soup = 5 and sandwiches = 8.
3. **Write the ratio** A student writes the ratio of soup to sandwiches as 8:5.
4. **Answer** The ratio of soup to sandwiches is 8:5.

Which step has the mistake? Explain what went wrong, then show the correct work.

> **HINT** Look at Step 3. Which amount should be written first — soup or sandwiches?

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What makes something a ratio? How did you decide which statements to sort into each group?

SENTENCE STARTER A ratio compares ___ to ___ using words like ___ or symbols like ___.

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.1

Understand Ratios

Version B · Core practice

MASTERY CHECK
- ☐ Exceeds
- ☐ Meets target
- ☐ Needs practice

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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

Sort each statement — is it a part-to-part ratio or a part-to-whole ratio?

Groups: Part-to-Part | Part-to-Whole

_____ 3 boys to 5 girls in cooking class

_____ 5 girls out of 8 students total

_____ 2 cups sugar to 3 cups flour

_____ 4 vegetarian dishes out of 10 total dishes

_____ 7 savory items to 3 sweet items

_____ 3 sweet items out of 10 total items

2 **Complete the table. Show how you found each value.**

Complete the table by writing each ratio in all three forms.

Description	Word Form	Colon Form	Fraction Form
4 cats to 7 dogs	4 to 7	4:7	4/7
9 red out of 15 total			
5 wins to 3 losses			

SHOW YOUR WORK

PART 2 Apply it

real-world problems

3 **Circle the letter of the best answer. Show how you know.**

A fruit salad has 5 pieces of melon, 3 pieces of pineapple, and 2 grapes. What is the ratio of pineapple to melon?

(A) 3:5
(B) 3:10
(C) 3:2
(D) 5:3

SHOW YOUR WORK

4 **Circle the letter of the best answer. Show how you know.**

A parking lot has 10 cars and 6 trucks. What is the part-to-whole ratio of trucks to total vehicles?

(A) 6:16
(B) 6:10
(C) 10:6
(D) 16:6

SHOW YOUR WORK

PART 3 Explain and correct

reasoning

5 **Find the mistake. Explain it, then show the correct work.**

Spot the Mistake — Part-to-Whole Ratio

1. **Read the problem** A class has 9 boys and 6 girls. Write the part-to-whole ratio of girls to the whole class.
2. **Count the whole** 9 + 6 = 15 students in all.
3. **Write the ratio** A student writes girls to the whole class as 6:9.
4. **Answer** The part-to-whole ratio of girls to the whole class is 6:9.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

EXPLAIN YOUR THINKING

What makes something a ratio? How did you decide which statements to sort into each group?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help

LESSON 3.1

Understand Ratios

Challenge · Extension

MASTERY CHECK
- ☐ Exceeds
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PART 1 Understand the idea

models and meaning

1 **Write the name of the correct group on each line.**

A classroom has 12 boys and 8 girls. Sort each ratio statement as either CORRECT or INCORRECT.

Groups: Correct Ratio | Incorrect Ratio

_____ Boys to girls = 12:8

_____ Girls to total = 8:20

_____ Boys to total = 12:8

_____ Girls to boys = 12:8

PART 2 Explain and correct

reasoning

2 **Find the mistake. Explain it, then show the correct work.**

Find Aliya's Mistake

1. **Read the problem** A trail mix has 6 almonds, 4 raisins, and 2 cashews. Write the ratio of raisins to total pieces.
2. **Count total pieces** 6 + 4 + 2 = 12
3. **Write the ratio** 4:8
4. **Answer** The ratio of raisins to total is 4:8.

Which step has the mistake? Explain what went wrong, then show the correct work.

CORRECT WORK

Correct answer: _____

3 **Answer in complete sentences.**

A bag of marbles has 8 red, 6 blue, and 4 green marbles. Write three different ratios using these quantities: one part-to-part ratio, one part-to-whole ratio, and explain the difference.

WRITE YOUR ANSWER

EXPLAIN YOUR THINKING

What makes something a ratio? How did you decide which statements to sort into each group?

3-Point Exit Checklist: ☐ 1. Units & Labels Verified ☐ 2. Mathematical Reasoning Stated ☐ 3. Calculations Double-Checked

How did it go? ☐ 4 · I can teach it ☐ 3 · I've got it ☐ 2 · I need a hint ☐ 1 · I need help